

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – Coding Challenge Task

Course Code:	CSA06	Course Title:	Design and Analysis of Algorithms
CO Assessed:	CO4: Articulation (BL4, BL5)	Assessment:	Assessment Tool 1 – Coding Challenge Task
Weightage:	40% of CIA	Total Marks:	100 (2 Questions × 50 Marks)
CO4 Statement:	Solve optimization problems using dynamic programming and greedy strategies.		
Student Name:	V Deepa Dharshini	Reg. No.:	192525213
Section / Batch:	B.Tech AIML(2 nd Year)	Date:	22-08-2026

Instructions to Students

- Answer both questions. Each question carries 50 Marks. Total: 100 Marks.
- Carefully analyze the given questions.
- Write clear code for the given problem.
- Analyze the Time complexity and Space complexity of your code using dynamic programming and greedy strategies

Question Type	Description	Questions	Total Marks
Coding Challenge Task	Focus on Code and analyze optimization problems using dynamic programming and greedy strategies	Q1 – Q2	100

RUBRICS FOR CALCULATION

Coding Challenge Task

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly understands problem constraints, edge cases, and competitive requirements	Understands problem with minor gaps in constraints	Partial understanding; misses key constraints	Misinterprets problem or constraints
Algorithm	25%	Selects optimal algorithm with best time and space complexity for competitive constraints	Uses efficient algorithm with minor scope for improvement	Uses basic/inefficient approach	No clear algorithmic approach
Code Implementation	20%	Writes correct, efficient, and clean code that passes all constraints	Code works with minor errors or inefficiencies	Partially correct code; fails for some cases	Code does not execute or gives incorrect results
Competitive Performance	20%	Solves problem within time limits and handles large inputs effectively	Solves within acceptable time with minor delays	Struggles with time limits or larger inputs	Unable to solve within time constraints
Test Case Handling	15%	Handles all test cases including edge and hidden cases	Handles most test cases	Misses important edge cases	Fails on multiple test cases

SECTION A – Coding Challenge Task

**SIMATS Online Programming Lab**
DAA PROGRAMMING

V Deepa Dharshini
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CO4-AT1-Coding Challenge

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Language: Python C C++ Java

QUESTIONS

Q1
Dynamic Programming
50 marks

Q2
Dynamic Programming
50 marks

2/2 answered

Q1 Dynamic Programming 50 marks

Unique Paths in Grid

A robot moves from the top-left corner to the bottom-right corner of a grid, moving only right or down. The goal is to determine the total number of unique paths. Efficient counting is required for large grids. Use Dynamic Programming to compute the result.

Input Format:
m, n = grid dimensions

Sample Input:
m = 3
n = 3

Expected Output:
Paths = 6

Your Code

```
1 #include <stdio.h>
2
3 int main() {
4     int m, n;
5     scanf("%d", &m);
6     scanf("%d", &n);
7
8     int dp[m][n];
9
10
11     for (int i = 0; i < m; i++)
12         dp[i][0] = 1;
13
14     for (int j = 0; j < n; j++)
15         dp[0][j] = 1;
16     for (int i = 1; i < m; i++) {
17         for (int j = 1; j < n; j++) {
18             dp[i][j] = dp[i-1][j] + dp[i][j-1];
19         }
20     }
21
22     printf("Paths = %d", dp[m-1][n-1]);
23
24     return 0;
25 }
```

Submitted

Input

```
3
3
```

Output

```
Paths = 6
```

**SIMATS Online Programming Lab**
DAA PROGRAMMING

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CO4-AT1-Coding Challenge

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Language: Python C C++ Java

QUESTIONS

Q1
Dynamic Programming
50 marks

Q2
Dynamic Programming
50 marks

2/2 answered

Q2 Dynamic Programming 50 marks

Maximum Product Subarray

Given an array of integers, find the contiguous subarray that has the maximum product. The array may contain negative numbers, which can affect the product value. The goal is to determine the maximum possible product. Use Dynamic Programming to compute the result efficiently.

Input Format:
n = number of elements
arr[] = array elements

Sample Input:
n = 5
arr = 2 3 -2 4 -1

Expected Output:
Max Product = 48

Your Code

```
1 #include <stdio.h>
2
3 long long max(long long a, long long b) {
4     return (a > b) ? a : b;
5 }
6
7 long long min(long long a, long long b) {
8     return (a < b) ? a : b;
9 }
10
11 int main() {
12     int n;
13     if (scanf("%d", &n) != 1 || n <= 0) return 0;
14
15     long long arr[n];
16     for (int i = 0; i < n; i++) {
17         scanf("%lld", &arr[i]);
18     }
19
20     long long max_prod = arr[0];
21     long long min_prod = arr[0];
22     long long result = arr[0];
23
24     for (int i = 1; i < n; i++) {
25         if (arr[i] < 0) {
26             long long temp = max_prod;
27             max_prod = min_prod;
```

Input

```
5
2 3 -2 4 -1
```

Output

```
Max Product = 48
```

Code of Conduct

I V Deepa Dharshini (192525213) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student:

**Mark Evaluation:**

Questions	Problem Understanding (20)	Algorithm (25)	Code Implementation (20)	Competitive Performance (20)	Test Case Handling (15)	Total Marks (Each - 50)
Q1						
Q2						
Overall Total						

Faculty In-charge	Course Coordinator	Head of Department
Name: _____ Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____